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Risk of aspiration pneumonia in pediatric patients with dysphagia who were found to have laryngeal penetration on the videofluoroscopic study. A systemic review protocol

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Title: Risk of aspiration pneumonia in pediatric patients with dysphagia who were found to have laryngeal penetration on the videofluoroscopic study. A systemic review protocol

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Abstract

Introduction

Dysphagia affects several children in the United States and around the globe. Videofluoroscopic swallow study (VFSS) is the most objective study to define swallowing function. Presence of tracheal aspiration during VFSS in children with dysphagia is associated with increased risk of aspiration pneumonia. However, the association of laryngeal penetration with aspiration pneumonia remains unclear. This systematic review aims to assess the risk of aspiration pneumonia in children with dysphagia with

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laryngeal penetration on VFSS and compare it with children who have tracheal aspiration and children who have neither tracheal aspiration nor laryngeal penetration.

Methods and Analysis

This study will be a systematic review and meta-analysis. Primary outcome will be the incidence of aspiration pneumonia. Secondary outcomes will include incidence of hospitalization, PICU admission, enteral tube requirement, growth, symptoms improvement, and mortality. The Cochrane Risk of Bias In Non-Randomized Studies of Interventions (ROBINS-I) tool will be used to assess the risk of bias in the included studies. Meta-analysis will be used to pool the studies. We will pool the dichotomous outcomes to obtain Odds ratio and report with a 95% confidence interval (CI). Continuous outcomes will be pooled to obtain mean difference and reported with a 95% CI. The overall grade of evidence will be assessed using GRADE criteria and findings will be presented in summary of findings table.

Ethics and Dissemination

This study is a systematic review without direct contact with patients. Therefore, IRB approval is not required. Authors consent for publishing this review. Data will be kept for review by editors and peer reviewers. Data will also be available to general public on request.

Registration: PROSPERO registration number CRD42020222145.

Article Summary

Strengths and limitations of this study

- An exhaustive electronic search on multiple databases will be utilized for systematic review and meta-analysis.
- The study will offer clarification about the risk of aspiration pneumonia in children who have laryngeal penetration on VFSS.
- We will use the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach to assess evidence quality.
- The studies included in the systematic review may not report all the outcomes of interest, which is a potential limitation.

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Main Manuscript

Introduction

About 500,000 children are affected by dysphagia in the United States[1]. Clinical evaluation of children with dysphagia involves obtaining a history, performing a physical examination, and additional studies such as a Videofluoroscopic swallow study (VFSS). The VFSS is the most objective study to evaluate the swallow function in a child. VFSS gives objective data about chewing, the progression of the bolus from the oral cavity to the pharynx, and information about laryngeal penetration and tracheal aspiration[2]. Laryngeal penetration is defined as the passage of material into the larynx without passing the vocal cords, whereas tracheal aspiration is defined as the passage of material below the vocal cords into the trachea[3]. The finding of tracheal aspiration on VFSS in children with dysphagia is associated with a three-fold increase in the risk of aspiration pneumonia[4]. Aspiration is among the most common diagnoses in infants who present with apparent life-threatening events[5]. While tracheal aspiration is strongly associated with aspiration pneumonia in children[6], the association of laryngeal penetration with aspiration pneumonia remains unclear[7, 8]. Some retrospective studies have shown an increased incidence of aspiration pneumonia in children with laryngeal penetration[9, 10], whereas others did not find a significant correlation[11, 12]. To the best of our knowledge, there are no studies that have systematically reviewed the published data on the association of laryngeal penetration and risk of aspiration pneumonia in children with dysphagia.

Objective

To assess the risk of aspiration pneumonia in children with dysphagia with laryngeal penetration on VFSS, and compare it with children who have tracheal aspiration and children who have neither tracheal aspiration nor laryngeal penetration.

Methods and analysis

We followed the PRISMA guidelines for reporting the protocol and will follow the same guidelines for reporting the main review. The protocol is registered at PROSPERO; registration number CRD42020222145.

Study type: We will include observational studies, including both cohort and case-control studies. We will exclude case series, case reports, and observational studies that did not have a comparison group.

Population: Population will include pediatric patients (<18 years) with dysphagia who have been evaluated with VFSS. We will exclude studies in which the VFSS was not done. We will exclude studies that only focused on patients with anatomical anomalies of the aerodigestive tract. We will, however, consider the studies that included a mixed population with and without anatomical problems.

Intervention/Exposure: Our primary exposure of interest is laryngeal penetration on VFSS in pediatric patients with dysphagia. The degree of laryngeal penetration will be categorized further as mild, moderate, and deep as follows:

Mild laryngeal penetration: Penetration of contrast medium under the tip of the epiglottis

Moderate laryngeal penetration: Presence of contrast medium in the upper two-thirds of the laryngeal vestibule

Deep laryngeal penetration: Presence of contrast medium into the lower one-third of the vestibule

Comparison: We will compare the exposure group with patients with tracheal aspiration, i.e., the passage of material below the vocal folds, and patients with neither penetration nor aspiration on VFSS. We will exclude studies that did not have a comparison group.

Outcomes:

Primary outcome

Incidence of aspiration pneumonia at 3 months, 6 months, and at the longest follow-up

We will use the WHO definition to define pneumonia in children between 2 to 59 months of age as cough and/or difficulty breathing with tachypnea and/or chest indrawing. 'Severe pneumonia' will be defined as pneumonia with general danger signs, including not being able to drink, persistent vomiting, convulsions, lethargy or unconsciousness, stridor in a calm child, or severe malnutrition[13].

Secondary outcomes

- Incidence of hospitalization at 3 months, 6 months, and at the longest follow-up
- Incidence of PICU admission at 3 months, 6 months, and at the longest follow-up
- Enteral tube requirement at 3 months, 6 months, and at the longest follow-up
- Mortality at 3 months, 6 months, and at the longest follow-up
- Weight for age at 3 months, 6 months, and at the longest follow-up
- Height for age at 3 months, 6 months, and at the longest follow-up
- BMI for age at 3 months, 6 months, and at the longest follow-up
- Changes in feeding practices (thickening or slowing) at 3 months, 6 months, and at the longest follow-up
-Symptoms improvement

Literature Search:

Systematic electronic searches will be conducted on multiple electronic databases, including PubMed, EMBASE, Web of Science, CINHAL, Scopus, Cochrane CENTRAL, LILACS, and WHO Global Index Medicus. We will also search the website clinicaltrial.gov to look for ongoing studies. No restrictions will be applied to the searches based on language, outcomes, publication status, or publication date. We will also search for the reference sections of previously published studies to look for eligible studies. We have two librarians on our team who will help with the literature searches. A proposed search strategy is shown in Appendix 1.

Data Extraction and Synthesis:

Selection of studies

The selection of studies will proceed by a three-stage process. Two authors will first screen the titles to identify potentially eligible studies. Any studies determined to be eligible at this step will proceed to the second stage of screening for a full-text review. Lastly, any studies retained after the full-text review will proceed to the third step of full data extraction. The software Covidence will be used to help with the screening and data extraction[14].

If a study is only available in abstract form, we will contact the authors to obtain complete details. If a study is available in another language, we will use local resources for translation. If a study is published in multiple publications, that study will be counted as one, and data will be extracted from all available sources.

Data Extraction

We will design and use a data extraction sheet to collect information from selected studies. Two authors (VA and LR) will independently use the data extraction sheets and compare their findings. Any incongruence or question will be resolved by discussion and with the help of the senior author on the team. We will extract the information for the study site, study year, study population, exposure (laryngeal penetration, tracheal aspiration), comparison, outcomes, risk of bias, and confounding factors.

The authors of the included studies may not follow a uniform definition of pneumonia. We will assess the definition of pneumonia in each of the included studies and see if they followed the WHO definition. If a study does not follow the WHO definition, we will note that but still include the data from that study as reported by the authors. Suppose the authors report the data on pneumonia based on severity (like mild, moderate, and severe), and these categories are mutually exclusive. In that case, we will combine these categories to make one outcome (Pneumonia, Yes or No).

Assessment of risk of bias in included studies

The Cochrane risk of bias in non-randomized studies of interventions (ROBINS-I) tool will be used to assess the risk of bias in studies. This tool will evaluate each study as a hypothetical randomized trial and will cover domains through which bias may be introduced. We will address five domains of "signaling questions" including: bias due to confounding, bias in the selection of participants into the study, bias due to missing data, bias in the measurement of outcomes, and bias in the selection of the reported result. Each domain will receive a judgment regarding the risk of bias – low, moderate, serious, or critical. The highest risk of bias in one domain will determine the overall risk of bias from a study, regardless of lower risks in other domains[15].

Data synthesis

Findings from all included studies will be reported in a narrative synthesis. We will also conduct meta-analyses to quantitatively synthesize evidence across studies if data are available from more than one study, and there is clinical and methodological homogeneity. Dichotomous outcomes will be combined to obtain Odd ratios and will be reported with 95% CI. Continuous outcomes will be combined using mean difference and will also be reported with their 95% CI. We will use the random effect models to pool the data. The software RevMAN[16] and STATA[17] will be used for statistical analysis.

We will include the occurrence of aspiration pneumonia, incidence of hospitalizations, PICU admission, enteral tube requirement, mortality, and changes in feeding practices as dichotomous values (yes/no). For these dichotomous outcomes, the total number of participants for each group, and the number of participants experiencing an event will be extracted. We will include weight for age, height for age, and BMI as continuous values. For continuous outcomes, the data will be pooled to obtain a mean difference. This will be reported with a 95% CI. If the data are reported in different units (e.g., some studies report weight in kg and others in Z scores), the standardized mean difference will be used and reported with a 95% CI. If a study does not report the standard deviation (SD) for a continuous outcome, we will contact the authors to obtain the data. If the authors do not have the SD data available, we will use SD from a similar study with a similar study population.

Dealing with missing data

Attrition will be noted during data extraction. We will contact authors if data is missing for some cases or if reasons for dropout are not reported. If a study does not report SD and we cannot retrieve these values from authors, we will find another study with a similar sample size that reports the same data

and uses the SD values reported in this study. If the data is missing for the key variables, we will write to the authors to request additional data.

Assessment of heterogeneity

Heterogeneity is defined as any variability among studies in a systematic review. Clinical heterogeneity is described as variability among participants, interventions, and outcomes studied. Methodological heterogeneity is variability in study design, outcome measurement tools, and risk of bias. Lastly, statistical heterogeneity is defined as the variability in the intervention effects being evaluated in different studies, resulting from clinical or methodological heterogeneity[18]. Statistical heterogeneity will be assessed using the I^2 statistics and χ^2 [19]. Low, moderate, and high levels of heterogeneity are defined with upper limits of 25%, 50%, and 75% for I^2, respectively. Calculated values are considered to be significant for heterogeneity when the I^2 value is >50% or the p-value is <0.1. Subgroup analyses will be performed to determine the reasons for any statistically significant heterogeneity.

Assessment of reporting bias

Small study effects and publication bias in meta-analysis will be interpreted with funnel plots. If funnel plot asymmetry is present, weighted linear regression (egger) tests will be utilized to determine the presence of bias when there are more than ten studies in the meta-analysis.

Subgroup analysis and investigation of heterogeneity

1) Infants (< 1 year of age) vs. children (1-5 years of age)
2) Children with CNS anomalies vs. children with no known CNS anomalies

Sensitivity analysis

1) Meta-analysis without studies with a high risk of bias
2) Random vs. fixed effect model

Rating of overall quality of evidence

Using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach with the help of software GradePro, we will assess the overall quality of evidence for the intervention's effect on each outcome[20]. This approach identifies and assesses many different features that affect the certainty of evidence in the review, including the type of study design, statistical heterogeneity, the directness of evidence, within-study risk of bias, risk of publication bias, and precision of effect estimates. Using GradePro, we will rate the overall quality of evidence as very low, low, moderate, or high. Observational studies begin as low-quality evidence; however, certain factors may lead to a rating up or down (Table 1).

Patient and Public Involvement:

There is no direct patient involvement in this study.

Ethics and dissemination:

Ethics approval and consent to participate

This study is a systematic review without direct contact with the patients. Therefore, IRB approval is not required for this study.

Consent for publication

Authors give consent for this review to be published.

Availability of data and materials

The entirety of the data will be kept for review by the editors and peer reviewers. The data will also be available to the general public on request.

Competing interests

Authors declare that they have no conflicts of interest.

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Authors' contributions

VA, LR, and AI designed the study and wrote the protocol. AS and OT helped develop search strategies. ETS reviewed the methodology. JM, ZS, AM, and RR are content experts and contributed accordingly.

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Table 1: GRADE method for rating up or down the quality of evidence [20]

Study Design	Quality of Evidence	Lower if	Higher if
Randomized trial →	High	Risk of bias -1 Serious -2 Very serious Inconsistency -1 Serious -2 Very serious Indirectness -1 Serious -2 Very serious Imprecision -1 Serious -2 Very serious Publication bias -1 Likely -2 Very likely	Large effect +1 Large +2 Very large Dose response +1 Evidence of a gradient All plausible confounding +1 Would reduce a demonstrated effect or +1 Would suggest a spurious effect when results show no effect
	Moderate		
Observational study →	Low		
	Very low		

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Appendix 1

Search strategies:

PubMed: 262 Results
((("Deglutition Disorders"[Mesh]) OR ((("deglutition" [tiab] AND "disorder*"[tiab]) OR ("deglutition"[tiab] AND "difficult*"[tiab]) OR ("swallow*"[tiab] AND "difficult*"[tiab]) OR ("swallow*"[tiab] AND "disorder*"[tiab]) OR "dysphagia*"[tiab] OR "aphagopraxia*"[tiab]))) AND ((((("videofluorographic*"[tiab] AND ("study"[tiab] OR "studies"[tiab] OR "exam*"[tiab] OR "swallow*"[tiab]))) OR (("videofluoroscopic*"[tiab] AND ("study"[tiab] OR "studies"[tiab] OR "exam*"[tiab] OR "swallow*"[tiab]))))) OR ("VFSS"[tiab] OR "VFSE"[tiab]))) AND ((((("Infant"[MeSH Terms]) OR ("Child"[MeSH Terms])) OR (Adolescent[MeSH Terms])) OR (("infant*"[tiab] OR "infancy"[tiab] OR "baby"[tiab] OR "babies"[tiab] OR "newborn*"[tiab] OR "newly born"[tiab] OR "new born*"[tiab] OR "neonat*"[tiab] OR "neo nat*"[tiab] OR "child*"[tiab] OR "toddler*"[tiab] OR "adolescen*"[tiab] OR "teen*"[tiab] OR "youth"[tiab] OR "juvenile*"[tiab])))
CINAHL: 25 Results
S1
(MH "Deglutition Disorders") OR TI (deglutition N2 disorder* OR deglutition N2 difficult* OR dysphagia* OR aphagopraxia* OR swallow* N2 difficult* OR swallow* N2 disorder*) OR AB (deglutition N2 disorder* OR deglutition N2 difficult* OR dysphagia* OR aphagopraxia* OR swallow* N2 difficult* OR swallow* N2 disorder*)
AND
S2
TI ((videofluoroscopic* N2 (study or studies OR swallow* OR exam*)) OR (videofluorographic* N2 (study or studies OR swallow* OR exam*)) OR (VFSS OR VFSE) OR AB ((videofluoroscopic* N2 (study or studies OR swallow* OR exam*)) OR (videofluorographic* N2 (study or studies OR swallow* OR exam*)) OR (VFSS OR VFSE)

AND
S3
TI ((infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR "neo nat*" OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*)) OR AB ((infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR "neo nat*" OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*)) OR ((MH "Infant+") OR (MH "Child+") OR (MH "Adolescence+"))
S1 AND S2 AND S3 Exclude MEDLINE records

Scopus: 22 Results
((TITLE-ABS (deglutition W/2 disorder*)) OR (TITLE-ABS (deglutition W/2 difficult*)) OR (TITLE-ABS (dysphagia* OR aphagopraxia*)) OR (TITLE-ABS (swallow* W/2 disorder*)) OR (TITLE-ABS (swallow* W/2 difficult*))) AND ((TITLE-ABS (videofluoroscopic* W/2 (swallow* OR study OR studies OR exam*))) OR (TITLE-ABS (videofluorographic* W/2 (swallow* OR study OR studies OR exam*))) OR (TITLE-ABS (vfss)) OR (TITLE-ABS (vfse))) AND (TITLE-ABS (infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR "neo nat*" OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*)) AND NOT INDEX (medline)
Embase: 125 results
#1 dysphagia'/exp
#2 'videofluoroscopic swallowing study'/exp

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#3 'videofluoroscopic swallow study'/exp
#4 'videofluorography'/exp
#5 'videofluoroscopy'/exp
#6 'examination'/exp
#7 'studies'/exp
#8 'study'/exp
#9 'swallowing'/exp
#10 'infant'/exp OR 'infancy'/exp OR 'baby'/exp OR 'newborn'/exp OR 'juvenile'/exp OR 'child'/exp OR 'adolescent'/exp OR 'adolescence'/exp
#11 dysphagia*:ti,ab OR aphagopraxia*:ti,ab
#12 (deglutition* NEAR/2 disorder*):ti,ab
#13 (deglutition* NEAR/2 difficult*):ti,ab
#14 (swallow* NEAR/2 disorder*):ti,ab
#15 (swallow* NEAR/2 difficult*):ti,ab
#16 #1 OR #11 OR #12 OR #13 OR #14 OR #15
#17 (videofluoroscopic* NEAR/2 (study OR studies OR exam* OR swallow*)):ti,ab
#18 (videofluorographic* NEAR/2 (study OR studies OR exam* OR swallow*)):ti,ab
#19 #6 OR #7 OR #8 OR #9
#20 #4 AND #19
#21 #5 AND #19
#22 #2 OR #3 OR #17 OR #20 OR #21
#23 infant*:ti,ab OR infancy:ti,ab OR baby:ti,ab OR babies:ti,ab OR newborn*:ti,ab OR 'new born*':ti,ab OR 'newly born':ti,ab OR neonat*:ti,ab OR 'neo nat*':ti,ab OR child*:ti,ab OR toddler*:ti,ab OR adolescen*:ti,ab OR teen*:ti,ab OR youth:ti,ab OR juvenile*:ti,ab
#24 #10 OR #23
#25 #16 AND #22 AND #24
#26 #25 NOT [medline]/lim

COCHRANE CENTRAL: 7 Results
#1 MeSH descriptor: [Deglutition Disorders] explode all trees
#2 deglutition NEAR disorder*:ti,ab OR deglutition NEAR difficult*:ti,ab OR swallow* NEAR disorder*:ti,ab OR swallow* NEAR difficult*:ti,ab OR dysphagia*:ti,ab OR aphagopraxia*:ti,ab
#3 VFSS:ti,ab OR VFSE:ti,ab
#4 (videofluoroscopic* NEAR (swallow* OR exam* OR study OR studies)):ti,ab
#5 (videofluorographic* NEAR (swallow* OR exam* OR study OR studies)):ti,ab
#6 MeSH descriptor: [Child] explode all trees
#7 MeSH descriptor: [Adolescent] explode all trees
#8 MeSH descriptor: [Infant] explode all trees
#9 infant*:ti,ab OR infancy:ti,ab OR baby:ti,ab OR babies:ti,ab OR newborn*:ti,ab OR new NEXT born*:ti,ab OR "newly born":ti,ab OR neonat*:ti,ab OR neo NEXT nat*:ti,ab OR child*:ti,ab OR toddler*:ti,ab OR adolescen*:ti,ab OR teen*:ti,ab OR youth:ti,ab OR juvenile*:ti,ab
#10 #1 OR #2
#11 #3 OR #4 OR #5
#12 #6 OR #7 OR #8 OR #9
#13 #10 AND #11 AND #12
#14 "accession number" near pubmed
#15 #13 NOT #14

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Web of Science: 230 Results
#1 TI= (deglutition NEAR/2 disorder*)
#2 TI= (deglutition NEAR/2 difficult*)
#3 TI= (swallow* NEAR/2 disorder*)
#4 TI= (swallow* NEAR/2 difficult*)
#5 TI= (dysphagia* OR aphagopraxia*)
#6 #5 OR #4 OR #3 OR #2 OR #1
#7 AB= (deglutition NEAR/2 disorder*)
#8 AB= (deglutition NEAR/2 difficult*)
#9 AB= (swallow* NEAR/2 disorder*)
#10 AB= (swallow* NEAR/2 difficult*)
#11 AB= (dysphagia* OR aphagopraxia*)
#12 #11 OR #10 OR #9 OR #8 OR #7
#13 #12 OR #6
#14 TI =(videofluoroscopic* NEAR/2 (swallow* OR study OR studies OR exam*))
#15 TI =(videofluorographic* NEAR/2 (swallow* OR study OR studies OR exam*))
#16 AB=(videofluorographic* NEAR/2 (swallow* OR study OR studies OR exam*))
#17 AB=(videofluoroscopic* NEAR/2 (swallow* OR study OR studies OR exam*))
#18 TI= (VFSS OR VFSE)
#19 AB= (VFSS OR VFSE)
#20 #18 OR #15 OR #14
#21 #19 OR #17 OR #16
#22 #21 OR #20
#23 TI=(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new* born*" OR toddler* OR adolescen* OR teen* OR child* OR youth OR juvenile*)
#24 AB=(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new* born*" OR toddler* OR adolescen* OR teen* OR child* OR youth OR juvenile*)
#25 #24 OR #23
#26 #25 AND #22 AND #13

LILACS: 14 Results – Global Index Medicus strategy built in LILACS VHL Platform. LILACS results will be exported as part of Global Index Medicus
(mh:("deglutition disorders")) OR ((ti:(deglutition AND disorder*)) OR ((ti:(swallow* AND disorder*))) OR ((ti:(dysphagia*))) OR ((ti:(aphagopraxia*)))) OR ((ab:(deglutition AND disorder*)) OR ((ab:(swallow* AND disorder*))) OR ((ab:(swallow* AND difficult*))) OR ((ab:(dysphagia*))) OR ((ab:(aphagopraxia*)))) AND ((ti:((videofluorographic* AND (study OR studies OR swallow*)))) OR ((ti:((videofluoroscopic* AND (study OR studies OR swallow*))))) OR ((ti:(VFSS OR VFSE))) OR ((ab:((videofluoroscopic* AND (study OR studies OR swallow* OR exam*)))) OR ((ab:((videofluorographic* AND (study OR studies OR swallow* OR exam*))))) OR ((ab:(VFSS OR VFSE))))) AND ((mh:("infant")) OR ((mh:("child"))) OR ((mh:("adolescent"))) OR ((ti:(infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*))) OR ((ab:(infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR "neo nat*" OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*))))

Global Index Medicus: 30 Results in Total: 16 Results WPRIM AND 14 Results LILACS
(mh:("deglutition disorders")) OR ((ti:(deglutition AND disorder*)) OR ((ti:(deglutition AND difficult*)) OR ((ti:(swallow* AND disorder*))) OR ((ti:(swallow* AND difficult*))) OR ((ti:(dysphagia*))) OR ((ti:(aphagopraxia*))))) OR ((ab:(deglutition AND disorder*)) OR ((ab:(deglutition AND difficult*)) OR

((ab:(swallow* AND disorder*))) OR ((ab:(swallow* AND difficult*))) OR ((ab:(dysphagia*))) OR ((ab:(aphagopraxia*))))) AND ((ti:((videofluorographic* AND (study OR studies OR swallow* OR exam*)))) OR ((ti:((videofluoroscopic* AND (study OR studies OR swallow* OR exam*))))) OR ((ti:(vfss OR vfse))) OR ((ab:((videofluoroscopic* AND (study OR studies OR swallow* OR exam*)))) OR ((ab:((videofluorographic* AND (study OR studies OR swallow* OR exam*))))) OR ((ab:(vfss OR vfse))))) AND ((mh:("infant")) OR ((mh:("child"))) OR ((mh:("adolescent"))) OR ((ti:(infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR “neo nat*” OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*))) OR ((ab:(infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR "neo nat*" OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*))))

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	X	☐	3
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	X	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	X	☐	114
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	X	☐	8-82
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	X	☐	328-330
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	X	
Support					
Sources	5a	Indicate sources of financial or other support for the review	X	☐	324-326
Sponsor	5b	Provide name for the review funder and/or sponsor	☐	X	
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐	X	
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	X	☐	128-144
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	X	☐	146-197

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	X	☐	146-214
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	X	☐	199-219
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	X	☐	Supplementary file
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	X	☐	221-234
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	X	☐	209-219
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	X	☐	221-234
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	X	☐	225-227
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	X	☐	179-197
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	X	☐	236-244
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	X	☐	246-263
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	X	☐	246-286
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	X	☐	288-294
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☐	X	

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	X	☐	283-286
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	X	☐	296-304

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BMJ Open

Risk of aspiration pneumonia in pediatric patients with dysphagia who were found to have laryngeal penetration on the instrumental swallow evaluation. A systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-048422.R1
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Primary Subject Heading:	Paediatrics
Secondary Subject Heading:	Gastroenterology and hepatology, Nutrition and metabolism
Keywords:	Paediatric gastroenterology < GASTROENTEROLOGY, Nutritional support < GASTROENTEROLOGY, Paediatric otolaryngology < OTOLARYNGOLOGY

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TITLE: Risk of aspiration pneumonia in pediatric patients with dysphagia who were found to have laryngeal penetration on the instrumental swallow evaluation. A systematic review protocol

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Key Words: Child, Deglutition Disorders, Pneumonia, Aspiration, Larynx

Word Count: 2693

ABSTRACT

Introduction

Dysphagia affects several children in United States and around the globe. Videofluoroscopic swallow study (VFSS) and fiberoptic endoscopic evaluation of swallowing (FEES) are the most objective studies to define swallowing function. Presence of tracheal aspiration during VFSS or FEES in children with dysphagia is associated with an increased risk of aspiration pneumonia. However, the association of laryngeal penetration with aspiration pneumonia remains unclear. This systematic review aims to assess the risk of aspiration pneumonia in children with dysphagia with laryngeal penetration on VFSS/FEES and compare it with children with tracheal aspiration and children with neither tracheal aspiration nor laryngeal penetration.

Methods and Analysis

This study will be a systematic review and meta-analysis. Systematic electronic searches will be conducted on PubMed, EMBASE, Web of Science, CINHAL, Scopus, Cochrane CENTRAL, LILACS,

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and WHO Global Index Medicus. We will include studies published through 6/10/2021. Primary outcome will be the incidence of aspiration pneumonia. Secondary outcomes will be incidence of hospitalization, PICU admission, enteral tube requirement, growth, symptoms improvement, and mortality. The Cochrane Risk of Bias In Non-Randomized Studies of Interventions (ROBINS-I) tool will be used to assess the risk of bias. Meta-analysis will be used to pool the studies. We will pool dichotomous outcomes to obtain an odds ratio and report with 95% confidence interval (CI). Continuous outcomes will be pooled to obtain mean difference and reported with 95% CI. Overall grade of evidence will be assessed using GRADE criteria, and findings will be presented in a summary of findings table.

Ethics and Dissemination

This study is a systematic review without contact with patients. Therefore, IRB approval is not required. Authors consent to publishing this review. Data will be kept for review by editors and peer reviewers. Data will be available to general public on request.

Registration: PROSPERO registration number CRD42020222145.

ARTICLE SUMMARY

Strengths and limitations of this study

- An exhaustive electronic search on multiple databases will be utilized for systematic review and meta-analysis.
- The study will offer clarification about the risk of aspiration pneumonia in children who have laryngeal penetration on Videofluoroscopic swallow study (VFSS) or fiberoptic endoscopic evaluation of swallowing (FEES).
- We will use the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach to assess evidence quality.
- The studies included in the systematic review may not report all the outcomes of interest, which is a potential limitation.

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MAIN MANUSCRIPT

Introduction

About 500,000 children are affected by dysphagia in the United States[1]. Clinical evaluation of children with dysphagia involves obtaining a history, performing a physical examination, and additional studies such as a Videofluoroscopic swallow study (VFSS) and fiberoptic endoscopic evaluation of swallowing (FEES). The VFSS is the most objective study to evaluate the swallow function in a child. VFSS gives objective data about chewing, the progression of the bolus from the oral cavity to the pharynx, and information about laryngeal penetration and tracheal aspiration[2]. Laryngeal penetration is defined as the passage of material into the larynx without passing the vocal cords. In contrast, tracheal aspiration is defined as the passage of material below the vocal cords into the trachea[3]. The finding of tracheal aspiration on VFSS in children with dysphagia is associated with a three-fold increase in the risk of aspiration pneumonia[4]. Aspiration is among the most common diagnoses in infants who present with apparent life-threatening events[5]. While tracheal aspiration is strongly associated with aspiration pneumonia in children[6], the association of laryngeal penetration with aspiration pneumonia remains unclear[7, 8]. Some retrospective studies have shown an increased incidence of aspiration pneumonia in children with laryngeal penetration[9, 10], whereas others did not find a significant correlation[11, 12]. To the best of our knowledge, there are no studies that have systematically reviewed the published data on the association of laryngeal penetration and the risk of aspiration pneumonia in children with dysphagia.

Objective

To assess the risk of aspiration pneumonia in children with dysphagia with laryngeal penetration on VFSS or FEES, compare it with children with tracheal aspiration and children who have neither tracheal aspiration nor laryngeal penetration.

Methods and analysis

We followed the PRISMA guidelines for reporting the protocol and will follow the same guidelines for reporting the main review. The protocol is registered at PROSPERO; registration number CRD42020222145.

Study type: We will include observational studies, including both cohort and case-control studies. We will exclude case series, case reports, and observational studies that did not have a comparison group.

Population: Population will include pediatric patients (<18 years) with dysphagia who have been evaluated with VFSS or Fiberoptic endoscopic evaluation of swallowing (FEES). We will consider studies that include a mixed population with and without anatomical problems. We will also include studies where the anatomical problem was not known before performing VFSS. However, we will exclude studies that focused exclusively on patients with a particular anatomical anomaly of the aerodigestive tract (e.g., cleft palate) or a particular medical diagnosis (e.g., Myasthenia Gravis, post-stroke); the results from those studies may not be generalizable to the population of interest.

Intervention/Exposure: Our primary exposure of interest is laryngeal penetration on VFSS or FEES in pediatric patients with dysphagia. We will exclude studies in which the VFSS or FEES was not done. We will consider exposure as yes or no (any laryngeal penetration vs. no laryngeal penetration for any consistency) and also consider the degrees of laryngeal penetration. The degree of laryngeal penetration will be categorized using the 8 point aspiration penetration scale as follows[3]:

1. Material does not enter airway.
2. Material enters the airway, remains above the vocal folds, and is ejected from the airway.
3. Material enters the airway, remains above the vocal folds, and is not ejected from the airway.

4. Material enters the airway, contacts the vocal folds, and is ejected from the airway.
5. Material enters the airway, contacts the vocal folds, and is not ejected from the airway.
6. Material enters the airway, passes below the vocal folds, and is ejected into the larynx or out of the airway.
7. Material enters the airway, passes below the vocal folds, and is not ejected from the trachea despite the effort.
8. Material enters the airway, passes below the vocal folds, and no effort is made to eject.

We will also consider categorizing the exposure based on mild/None vs. moderate vs. deep laryngeal penetration. The definition of none/mild vs. Moderate vs. deep laryngeal penetration will be based on definitions given in reference[8]: None/Mild laryngeal penetration: penetration of contrast medium under the tip of the epiglottis; Moderate penetration: entry of contrast medium into the upper two-thirds of the laryngeal vestibule and remaining above the arbitrary plane that extends from the top of the arytenoids to the top of the thyroid cartilage and; Deep laryngeal penetration: entry of contrast medium into the lower one-third of the laryngeal vestibule below the arbitrary plane that extends from the top of the arytenoids to the top of the thyroid cartilage; this level of laryngeal penetration may or may not result in the coating of the vocal cords. We will consider the penetration score discussed above: Score 0-3 as mild/none, score 4: Moderate and score 5: deep laryngeal penetration.

Comparison: We will compare the exposure group with patients with tracheal aspiration, i.e., the passage of material below the vocal folds, and patients with neither penetration nor aspiration on VFSS. We will exclude studies that did not have a comparison group.

Outcomes:

Primary outcome

Incidence of aspiration pneumonia (any severity): Dichotomous outcome measured at 3 months, 6 months, and at the longest follow-up

We will use the WHO definition to define pneumonia in children between 2 to 59 months of age as cough and/or difficulty breathing with tachypnea and/or chest indrawing. 'Severe pneumonia' will be defined as pneumonia with general danger signs, including not being able to drink, persistent vomiting, convulsions, lethargy or unconsciousness, stridor in a calm child, or severe malnutrition[13].

Secondary outcomes

- Incidence of hospitalization: Dichotomous outcome measured at 3 months, 6 months, and at the longest follow-up
- Incidence of pediatric intensive care unit (PICU) admission: Dichotomous outcome measured at 3 months, 6 months, and at the longest follow-up
- Enteral tube requirement: Dichotomous outcome measured at 3 months, 6 months, and at the longest follow-up
- Mortality: Dichotomous outcome measured at 3 months, 6 months, and at the longest follow-up
- Weight for age (kg or Z scores): Continous outcome measured at 3 months, 6 months, and at the longest follow-up
- Height for age (cm or Z scores): Continous outcome measured at 3 months, 6 months, and at the longest follow-up
- BMI for age (Kg/m^2 or z scores): Continous outcome measured at 3 months, 6 months, and at the longest follow-up
- Changes in feeding practices: (thickening or slowing) Dichotomus outcome measured at 3 months, 6 months, and at the longest follow-up

-Adverse events: Dichotomous outcome

Literature Search:
Systematic electronic searches will be conducted on multiple electronic databases, including PubMed, EMBASE, Web of Science, CINHAL, Scopus, Cochrane CENTRAL, LILACS, and WHO Global Index Medicus. We will include studies published through 6/10/2021. We will also search the website clinicaltrial.gov to look for ongoing studies. No restrictions will be applied to the searches based on language, outcomes, publication status, or publication date. We will also search for the reference sections of previously published studies to look for eligible studies. We have two librarians on our team who will help with the literature searches. A proposed search strategy is shown in Appendix 1.

Data Extraction and Synthesis:
Selection of studies
The selection of studies will proceed by a three-stage process. Two authors will first screen the titles to identify potentially eligible studies. Any studies determined to be eligible at this step will proceed to the second stage of screening for a full-text review. Lastly, any studies retained after the full-text review will proceed to the third step of full data extraction. The software Covidence will be used to help with the screening and data extraction[14].

If a study is only available in abstract form, we will contact the authors to obtain complete details. If a study is available in another language, we will use local resources for translation. If a study is published in multiple publications, that study will be counted as one, and data will be extracted from all available sources.

Data Extraction
We will design and use a data extraction sheet to collect information from selected studies. Two authors (VA and LR) will independently use the data extraction sheets and compare their findings. Any incongruence or question will be resolved by discussion and with the help of the senior author on the team. We will extract the information for the study site, study year, study population, exposure (laryngeal penetration, tracheal aspiration), comparison, outcomes, risk of bias, and confounding factors.

The authors of the included studies may not follow a uniform definition of pneumonia. We will assess the definition of pneumonia in each of the included studies and see if they followed the WHO definition. If a study does not follow the WHO definition, we will note that but still include the data from that study as reported by the authors. If the authors report the data on pneumonia based on severity (like mild, moderate, and severe), and these categories are mutually exclusive. In that case, we will combine these categories to make one outcome (Pneumonia, Yes or No).

Assessment of risk of bias in included studies:
The Cochrane risk of bias in non-randomized studies of interventions (ROBINS-I) tool will be used to assess the risk of bias in studies. This tool will evaluate each study as a hypothetical randomized trial and will cover domains through which bias may be introduced. We will address five domains of "signaling questions," including bias due to confounding, bias in the selection of participants into the study, bias due to missing data, bias in the measurement of outcomes, and bias in selecting the reported result. Each domain will receive a judgment regarding the risk of bias – low, moderate, serious, or critical. The highest risk of bias in one domain will determine the overall risk of bias from a study, regardless of lower risks in other domains[15].

Data synthesis:
Findings from all included studies will be reported in a narrative synthesis. We will also conduct meta-analyses to quantitatively synthesize evidence across studies if data are available from more than one study and there is clinical and methodological homogeneity. Dichotomous outcomes will be combined to

obtain Odd ratios and will be reported with 95% CI. Continuous outcomes will be combined using mean difference and will also be reported with their 95% CI. We will use the random effect models to pool the data using the generic inverse variance method of meta-analysis. The software RevMAN[16] and STATA[17] will be used for statistical analysis.

We will include the occurrence of aspiration pneumonia, incidence of hospitalizations, PICU admission, enteral tube requirement, mortality, and changes in feeding practices as dichotomous values and adverse events (yes/no). For these dichotomous outcomes, the total number of participants for each group and the number of participants experiencing an event will be extracted. In addition, we will include weight for age, height for age, and BMI as continuous values. For continuous outcomes, the data will be pooled to obtain a mean difference. This will be reported with a 95% CI. If the data are reported in different units (e.g., some studies report weight in kg and others in Z scores), the standardized mean difference will be used and reported with a 95% CI. If a study does not report the standard deviation (SD) for a continuous outcome, we will contact the authors to obtain the data. If the authors do not have the SD data available, we will use SD from a similar study with a similar study population.

Dealing with missing data:

Attrition will be noted during data extraction. We will contact authors if data is missing for some cases or if reasons for dropout are not reported. If a study does not report SD and we cannot retrieve these values from authors, we will find another study with a similar sample size that reports the same data and uses the SD values reported in this study. If the data is missing for the key variables, we will write to the authors to request additional data.

Assessment of heterogeneity:

Heterogeneity is defined as any variability among studies in a systematic review. Clinical heterogeneity is described as variability among participants, interventions, and outcomes studied. Methodological heterogeneity is variability in study design, outcome measurement tools, and risk of bias. Lastly, statistical heterogeneity is defined as the variability in the intervention effects being evaluated in different studies, resulting from clinical or methodological heterogeneity[18]. Statistical heterogeneity will be assessed using the I^2 statistics and χ^2 [19]. Low, moderate, and high levels of heterogeneity are defined with upper limits of 25%, 50%, and 75% for I^2, respectively. Calculated values are considered to be significant for heterogeneity when the I^2 value is >50% or the p-value is <0.1. Subgroup analyses will be performed to determine the reasons for any statistically significant heterogeneity.

Assessment of reporting bias:

Small study effects and publication bias in meta-analysis will be interpreted with funnel plots. If funnel plot asymmetry is present, weighted linear regression (egger) tests will be utilized to determine the presence of bias when there are more than ten studies in the meta-analysis.

Subgroup analysis and investigation of heterogeneity

1) Study population: Age: Infants (< 1 year of age) vs. children (1-5 years of age)
2) Study population: Neurological comorbidity: Children with CNS anomalies vs. children with no known CNS anomalies
3) Study population: Syndromic comorbidity: Children with a syndromic diagnosis vs. children with no syndromic diagnosis
4) Study population: Anatomic anomalies of airway or gastrointestinal tract: Children with anatomical anomalies vs. children with no anatomical anomalies.
5) Exposure: Aspiration scale score: Scores of 0-2 vs. 3-5 vs. 6-8
6) Exposure: Laryngeal penetration: None/mild vs. Moderate vs. Deep penetration

Exposure: Consistency of food: Thin liquid vs. thick liquid vs. Puree vs. solid food
Exposure: Severity of impairment of swallow reflex initi[8]ation: Normal vs. Mild, moderate vs. severe vs. Profound

We will use the following definitions for the severity of impairment of swallow reflex initiation[8]:

Normal: Swallow reflex initiated at the back of the tongue with no hesitation as the bolus moves from the back of the tongue into the pharynx

Mild: Swallow reflex initiated in the mid-pharynx (vallecular spaces) after a brief hesitation

Moderate: Swallow reflex initiated in the lower pharynx (pyriform sinuses) after a brief hesitation

Severe: Swallow Reflex initiated in the lower pharynx (pyriform sinuses) after prolonged pooling

Profound: Absent swallow reflex

These subgroup analyses will be considered based on subgroup effects reported in studies based on individual patient features

Sensitivity analysis

1) Meta-analysis without studies with a high risk of bias
2) Random vs. fixed effect model
3) WHO definition of pneumonia vs. non-WHO/non-standard definition of pneumonia

Rating of overall quality of evidence

Using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach with the help of the software GradePro, we will assess the overall quality of evidence for the intervention's effect on each outcome[20]. This approach identifies and assesses many different features that affect the certainty of evidence in the review, including the type of study design, statistical heterogeneity, the directness of evidence, within-study risk of bias, risk of publication bias, and precision of effect estimates. Using GradePro, we will rate the overall quality of evidence as very low, low, moderate, or high. Observational studies begin as low-quality evidence; however, certain factors may lead to a rating up or down (Table 1).

Patient and Public Involvement

There is no direct patient involvement in this study.

Ethics and dissemination:

Ethics approval and consent to participate

This study is a systematic review without direct contact with the patients. Therefore, IRB approval is not required for this study.

Consent for publication

Authors give consent for this review to be published.

Availability of data and materials

The entirety of the data will be kept for review by the editors and peer reviewers. The data will also be available to the general public on request.

Competing interests
The authors declare that they have no conflicts of interest.

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Authors' contributions
VA, LR, and AI designed the study and wrote the protocol. AS and OT helped develop search strategies. ETS reviewed the methodology. JM, ZS, AM, and RR are content experts and contributed accordingly.

Acknowledgments
None

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Table 1: GRADE method for rating up or down the quality of evidence [20]

Study Design	Quality of Evidence	Lower if	Higher if
Randomized trial →	High	Risk of bias -1 Serious -2 Very serious Inconsistency -1 Serious -2 Very serious Indirectness -1 Serious -2 Very serious Imprecision -1 Serious -2 Very serious Publication bias -1 Likely -2 Very likely	Large effect +1 Large +2 Very large Dose response +1 Evidence of a gradient All plausible confounding +1 Would reduce a demonstrated effect or +1 Would suggest a spurious effect when results show no effect
	Moderate		
Observational study →	Low		
	Very low		

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Appendix 1

Search strategies:

PubMed: 262 Results
((("Deglutition Disorders"[Mesh]) OR ((("deglutition" [tiab] AND "disorder*"[tiab]) OR ("deglutition"[tiab] AND "difficult*"[tiab]) OR ("swallow*"[tiab] AND "difficult*"[tiab]) OR ("swallow*"[tiab] AND "disorder*"[tiab]) OR "dysphagia*"[tiab] OR "aphagopraxia*"[tiab]))) AND ((((("videofluorographic*"[tiab] AND ("study"[tiab] OR "studies"[tiab] OR "exam*"[tiab] OR "swallow*"[tiab]))) OR (("videofluoroscopic*"[tiab] AND ("study"[tiab] OR "studies"[tiab] OR "exam*"[tiab] OR "swallow*"[tiab]))))) OR ("VFSS"[tiab] OR "VFSE"[tiab]))) AND ((((("Infant"[MeSH Terms]) OR ("Child"[MeSH Terms])) OR (Adolescent[MeSH Terms])) OR (("infant*"[tiab] OR "infancy"[tiab] OR "baby"[tiab] OR "babies"[tiab] OR "newborn*"[tiab] OR "newly born"[tiab] OR "new born*"[tiab] OR "neonat*"[tiab] OR "neo nat*"[tiab] OR "child*"[tiab] OR "toddler*"[tiab] OR "adolescen*"[tiab] OR "teen*"[tiab] OR "youth"[tiab] OR "juvenile*"[tiab])))
CINAHL: 25 Results
S1
(MH "Deglutition Disorders") OR TI (deglutition N2 disorder* OR deglutition N2 difficult* OR dysphagia* OR aphagopraxia* OR swallow* N2 difficult* OR swallow* N2 disorder*) OR AB (deglutition N2 disorder* OR deglutition N2 difficult* OR dysphagia* OR aphagopraxia* OR swallow* N2 difficult* OR swallow* N2 disorder*)
AND
S2
TI ((videofluoroscopic* N2 (study or studies OR swallow* OR exam*)) OR (videofluorographic* N2 (study or studies OR swallow* OR exam*)) OR (VFSS OR VFSE) OR AB ((videofluoroscopic* N2 (study or studies OR swallow* OR exam*)) OR (videofluorographic* N2 (study or studies OR swallow* OR exam*)) OR (VFSS OR VFSE)

AND
S3
TI ((infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR "neo nat*" OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*)) OR AB ((infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR "neo nat*" OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*)) OR ((MH "Infant+") OR (MH "Child+") OR (MH "Adolescence+"))
S1 AND S2 AND S3 Exclude MEDLINE records

Scopus: 22 Results
((TITLE-ABS (deglutition W/2 disorder*)) OR (TITLE-ABS (deglutition W/2 difficult*)) OR (TITLE-ABS (dysphagia* OR aphagopraxia*)) OR (TITLE-ABS (swallow* W/2 disorder*)) OR (TITLE-ABS (swallow* W/2 difficult*))) AND ((TITLE-ABS (videofluoroscopic* W/2 (swallow* OR study OR studies OR exam*))) OR (TITLE-ABS (videofluorographic* W/2 (swallow* OR study OR studies OR exam*))) OR (TITLE-ABS (vfss)) OR (TITLE-ABS (vfse))) AND (TITLE-ABS (infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR "neo nat*" OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*)) AND NOT INDEX (medline)
Embase: 125 results
#1 dysphagia'/exp
#2 'videofluoroscopic swallowing study'/exp

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#3 'videofluoroscopic swallow study'/exp
#4 'videofluorography'/exp
#5 'videofluoroscopy'/exp
#6 'examination'/exp
#7 'studies'/exp
#8 'study'/exp
#9 'swallowing'/exp
#10 'infant'/exp OR 'infancy'/exp OR 'baby'/exp OR 'newborn'/exp OR 'juvenile'/exp OR 'child'/exp OR 'adolescent'/exp OR 'adolescence'/exp
#11 dysphagia*:ti,ab OR aphagopraxia*:ti,ab
#12 (deglutition* NEAR/2 disorder*):ti,ab
#13 (deglutition* NEAR/2 difficult*):ti,ab
#14 (swallow* NEAR/2 disorder*):ti,ab
#15 (swallow* NEAR/2 difficult*):ti,ab
#16 #1 OR #11 OR #12 OR #13 OR #14 OR #15
#17 (videofluoroscopic* NEAR/2 (study OR studies OR exam* OR swallow*)):ti,ab
#18 (videofluorographic* NEAR/2 (study OR studies OR exam* OR swallow*)):ti,ab
#19 #6 OR #7 OR #8 OR #9
#20 #4 AND #19
#21 #5 AND #19
#22 #2 OR #3 OR #17 OR #20 OR #21
#23 infant*:ti,ab OR infancy:ti,ab OR baby:ti,ab OR babies:ti,ab OR newborn*:ti,ab OR 'new born*':ti,ab OR 'newly born':ti,ab OR neonat*:ti,ab OR 'neo nat*':ti,ab OR child*:ti,ab OR toddler*:ti,ab OR adolescen*:ti,ab OR teen*:ti,ab OR youth:ti,ab OR juvenile*:ti,ab
#24 #10 OR #23
#25 #16 AND #22 AND #24
#26 #25 NOT [medline]/lim

COCHRANE CENTRAL: 7 Results
#1 MeSH descriptor: [Deglutition Disorders] explode all trees
#2 deglutition NEAR disorder*:ti,ab OR deglutition NEAR difficult*:ti,ab OR swallow* NEAR disorder*:ti,ab OR swallow* NEAR difficult*:ti,ab OR dysphagia*:ti,ab OR aphagopraxia*:ti,ab
#3 VFSS:ti,ab OR VFSE:ti,ab
#4 (videofluoroscopic* NEAR (swallow* OR exam* OR study OR studies)):ti,ab
#5 (videofluorographic* NEAR (swallow* OR exam* OR study OR studies)):ti,ab
#6 MeSH descriptor: [Child] explode all trees
#7 MeSH descriptor: [Adolescent] explode all trees
#8 MeSH descriptor: [Infant] explode all trees
#9 infant*:ti,ab OR infancy:ti,ab OR baby:ti,ab OR babies:ti,ab OR newborn*:ti,ab OR new NEXT born*:ti,ab OR "newly born":ti,ab OR neonat*:ti,ab OR neo NEXT nat*:ti,ab OR child*:ti,ab OR toddler*:ti,ab OR adolescen*:ti,ab OR teen*:ti,ab OR youth:ti,ab OR juvenile*:ti,ab
#10 #1 OR #2
#11 #3 OR #4 OR #5
#12 #6 OR #7 OR #8 OR #9
#13 #10 AND #11 AND #12
#14 "accession number" near pubmed
#15 #13 NOT #14

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Web of Science: 230 Results
#1 TI= (deglutition NEAR/2 disorder*)
#2 TI= (deglutition NEAR/2 difficult*)
#3 TI= (swallow* NEAR/2 disorder*)
#4 TI= (swallow* NEAR/2 difficult*)
#5 TI= (dysphagia* OR aphagopraxia*)
#6 #5 OR #4 OR #3 OR #2 OR #1
#7 AB= (deglutition NEAR/2 disorder*)
#8 AB= (deglutition NEAR/2 difficult*)
#9 AB= (swallow* NEAR/2 disorder*)
#10 AB= (swallow* NEAR/2 difficult*)
#11 AB= (dysphagia* OR aphagopraxia*)
#12 #11 OR #10 OR #9 OR #8 OR #7
#13 #12 OR #6
#14 TI =(videofluoroscopic* NEAR/2 (swallow* OR study OR studies OR exam*))
#15 TI =(videofluorographic* NEAR/2 (swallow* OR study OR studies OR exam*))
#16 AB=(videofluorographic* NEAR/2 (swallow* OR study OR studies OR exam*))
#17 AB=(videofluoroscopic* NEAR/2 (swallow* OR study OR studies OR exam*))
#18 TI= (VFSS OR VFSE)
#19 AB= (VFSS OR VFSE)
#20 #18 OR #15 OR #14
#21 #19 OR #17 OR #16
#22 #21 OR #20
#23 TI=(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new* born*" OR toddler* OR adolescen* OR teen* OR child* OR youth OR juvenile*)
#24 AB=(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new* born*" OR toddler* OR adolescen* OR teen* OR child* OR youth OR juvenile*)
#25 #24 OR #23
#26 #25 AND #22 AND #13

LILACS: 14 Results – Global Index Medicus strategy built in LILACS VHL Platform. LILACS results will be exported as part of Global Index Medicus
(mh:("deglutition disorders")) OR ((ti:(deglutition AND disorder*)) OR ((ti:(swallow* AND disorder*))) OR ((ti:(dysphagia*))) OR ((ti:(aphagopraxia*)))) OR ((ab:(deglutition AND disorder*)) OR ((ab:(swallow* AND disorder*))) OR ((ab:(swallow* AND difficult*))) OR ((ab:(dysphagia*))) OR ((ab:(aphagopraxia*)))) AND ((ti:((videofluorographic* AND (study OR studies OR swallow*)))) OR ((ti:((videofluoroscopic* AND (study OR studies OR swallow*))))) OR ((ti:(VFSS OR VFSE))) OR ((ab:((videofluoroscopic* AND (study OR studies OR swallow* OR exam*)))) OR ((ab:((videofluorographic* AND (study OR studies OR swallow* OR exam*))))) OR ((ab:(VFSS OR VFSE))))) AND ((mh:("infant")) OR ((mh:("child"))) OR ((mh:("adolescent"))) OR ((ti:(infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*))) OR ((ab:(infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR "neo nat*" OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*))))

Global Index Medicus: 30 Results in Total: 16 Results WPRIM AND 14 Results LILACS
(mh:("deglutition disorders")) OR ((ti:(deglutition AND disorder*)) OR ((ti:(deglutition AND difficult*)) OR ((ti:(swallow* AND disorder*))) OR ((ti:(swallow* AND difficult*))) OR ((ti:(dysphagia*))) OR ((ti:(aphagopraxia*))))) OR ((ab:(deglutition AND disorder*)) OR ((ab:(deglutition AND difficult*)) OR

((ab:(swallow* AND disorder*))) OR ((ab:(swallow* AND difficult*))) OR ((ab:(dysphagia*))) OR ((ab:(aphagopraxia*))))) AND ((ti:((videofluorographic* AND (study OR studies OR swallow* OR exam*)))) OR ((ti:((videofluoroscopic* AND (study OR studies OR swallow* OR exam*))))) OR ((ti:(vfss OR vfse))) OR ((ab:((videofluoroscopic* AND (study OR studies OR swallow* OR exam*)))) OR ((ab:((videofluorographic* AND (study OR studies OR swallow* OR exam*))))) OR ((ab:(vfss OR vfse))))) AND ((mh:("infant")) OR ((mh:("child"))) OR ((mh:("adolescent"))) OR ((ti:(infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR “neo nat*” OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*))) OR ((ab:(infant* OR infancy OR baby OR babies OR newborn* OR "new born*" OR "newly born" OR neonat* OR "neo nat*" OR child* OR toddler* OR adolescen* OR teen* OR youth OR juvenile*))))

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	X	☐	3
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	X	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	X	☐	114
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	X	☐	8-82
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	X	☐	328-330
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	X	
Support					
Sources	5a	Indicate sources of financial or other support for the review	X	☐	324-326
Sponsor	5b	Provide name for the review funder and/or sponsor	☐	X	
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐	X	
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	X	☐	128-144
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	X	☐	146-197

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	X	☐	146-214
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	X	☐	199-219
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	X	☐	Supplementary file
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	X	☐	221-234
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	X	☐	209-219
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	X	☐	221-234
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	X	☐	225-227
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	X	☐	179-197
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	X	☐	236-244
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	X	☐	246-263
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	X	☐	246-286
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	X	☐	288-294
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☐	X	

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	X	☐	283-286
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	X	☐	296-304

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